



PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2014

**Candidate
Name**

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**Civics
Group**

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NUMBER**

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H2 BIOLOGY

9648 / 01

PAPER 1 Multiple Choice

25 Sept 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the cover page in the spaces provided unless this has been done for you.

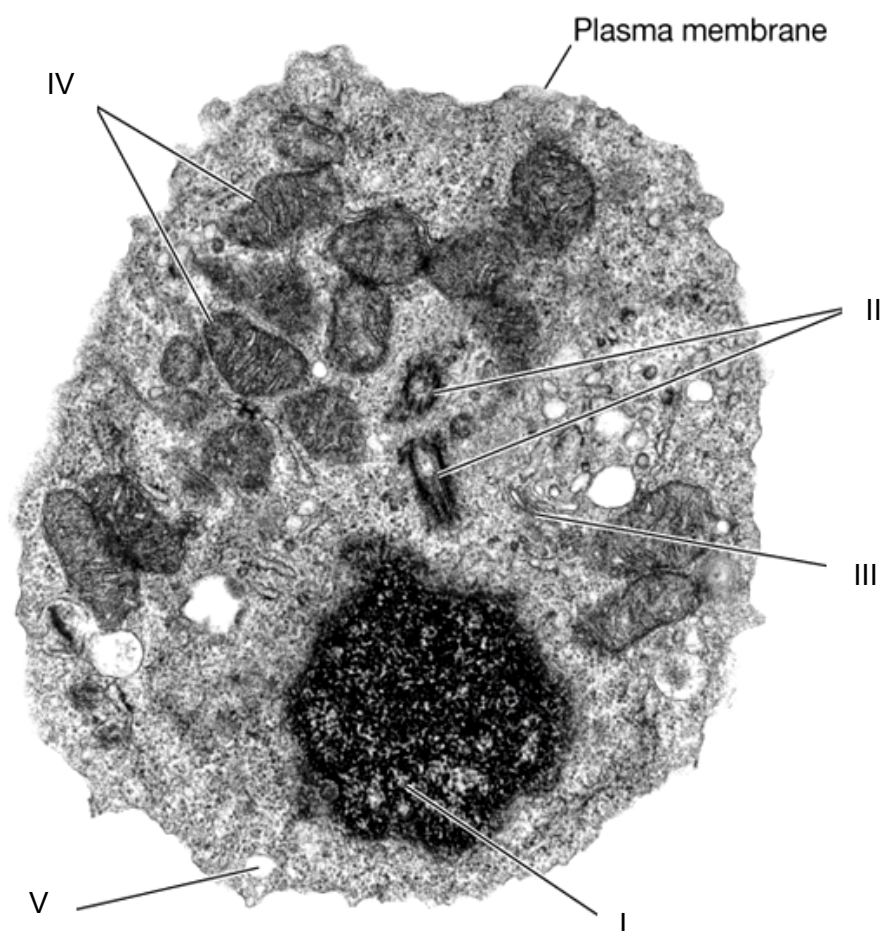
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OTAS answer sheet provided.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Do not open this booklet until you are told to do so.

- 1 An electron micrograph of a cell is shown below.



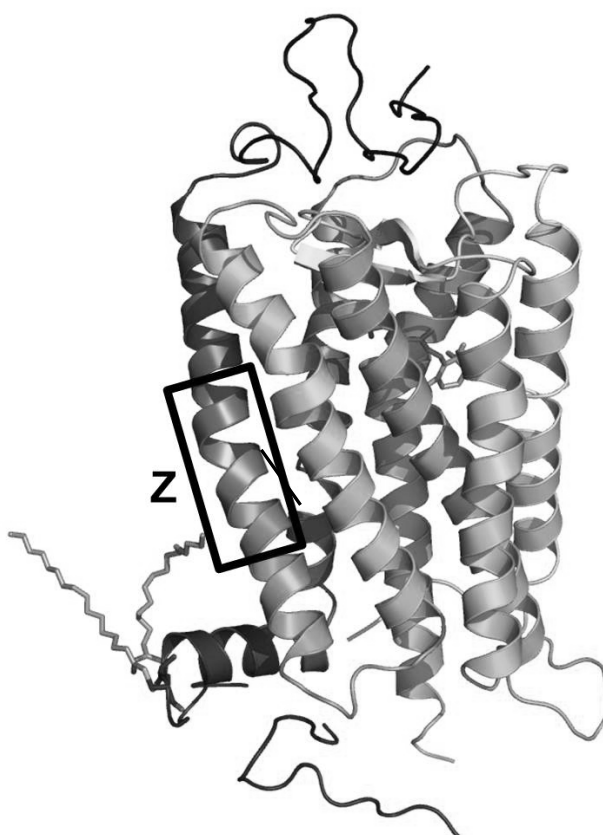
Match the organelles I, II, III, IV and V associated with the cellular processes listed.

	I	II	III	IV	V
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

2 Which combination would decrease the fluidity of a cell surface membrane?

	Saturated fatty acids	Non-saturated fatty acids	Cholesterol
A	Decrease	Increase	Decrease
B	Decrease	Increase	Increase
C	Increase	Decrease	Decrease
D	Increase	Decrease	Increase

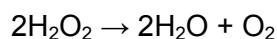
3 The diagram below shows the structure of a rhodopsin found in animal eyes.



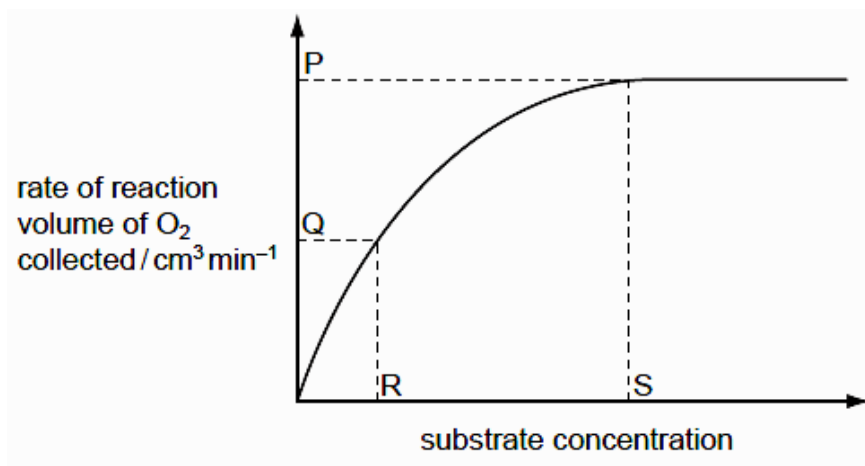
Which correctly describes the structure of the part enclosed in box Z?

	number of amino acids present	types of bonds present
A	cannot be determined	hydrogen bonds only
B	cannot be determined	hydrogen bonds and hydrophilic bonds
C	approximately 11 amino acids	hydrogen bonds only
D	approximately 11 amino acids	hydrogen bonds and hydrophobic bonds

- 4 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.



The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time. Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

- I. At P, the rate of reaction is limited by the concentration of enzyme.
 - II. At Q, all of the enzyme active sites are occupied by substrate molecules.
 - III. At Q, the rate of reaction is limited by the concentration of the substrate.
 - IV. At S, all of the enzyme active sites are occupied by substrate molecules.
- A** I and IV
B II and III
C II and IV
D I, III and IV

- 5 The diagram shows the chromosomes of a cell which has been squashed during mitosis.



Which of the following correctly describes the chromosomes and cell shown?

	Stage of cell division	Number of chromosomes	Type of cell	Total number of DNA strands
A	anaphase	5	Haploid	20
B	metaphase	5	Diploid	10
C	anaphase	10	Haploid	40
D	metaphase	10	Diploid	40

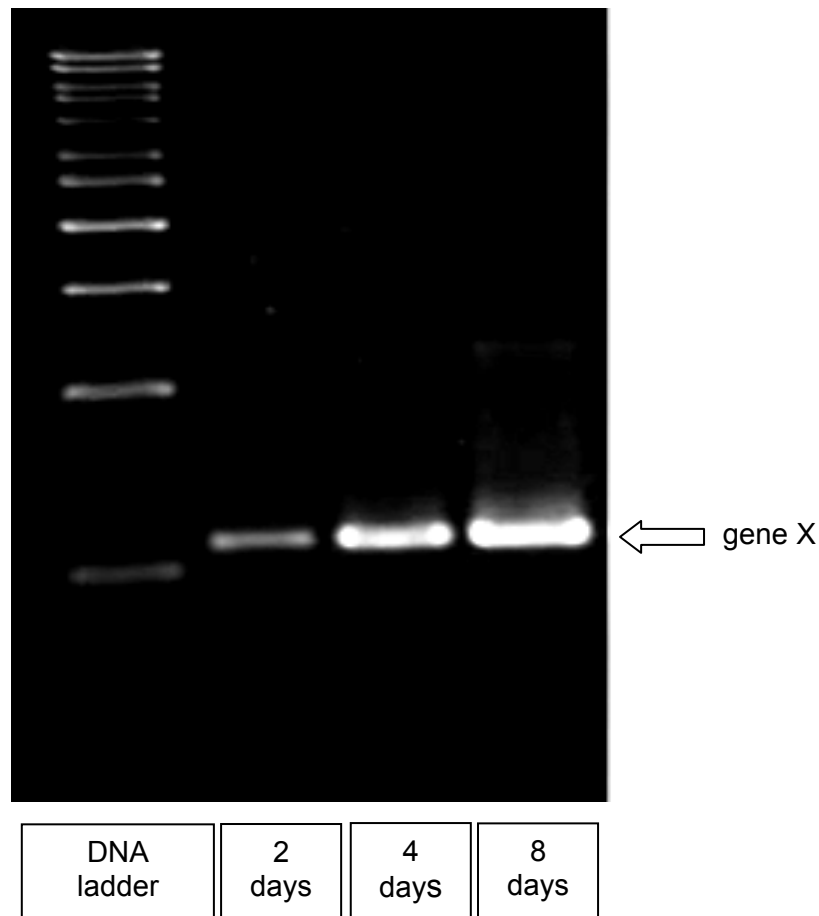
- 6 A student obtained a sample of DNA molecule. mRNA was transcribed from this DNA molecule. He then separated the two strands of the DNA sample by adding NaOH. The base compositions of each strand, that of the mRNA and a foreign DNA strand were analysed. The results of the analysis are shown in the table below.

	A	G	C	T	U
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

Which strand of DNA serves as a template for mRNA synthesis?

- A** Strand 1
- B** Strand 2
- C** Strand 3
- D** Strand 2 and 3

- 7 Gel electrophoresis was performed using samples of gene X isolated from equal number of cells from a human embryo after 2 days, 4 days and 8 days of development.

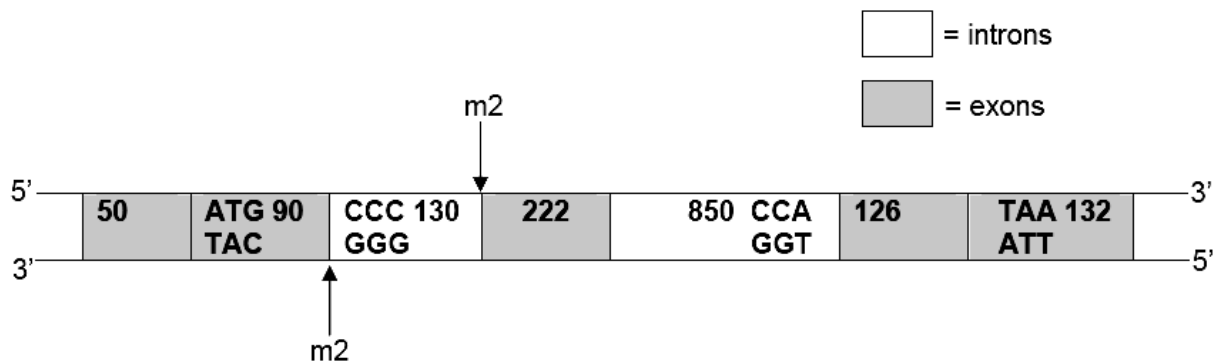


What kind of gene regulation is illustrated by the results of gel electrophoresis?

- A DNA methylation
- B gene amplification
- C histone acetylation
- D transcriptional activation

Use the information below to answer Questions 8 and 9.

The figure below shows the genomic structure of the wild-type human β -globin gene. The numbers within the boxes indicate the length of nucleotides of each region, **inclusive** of bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are indicated.



- 8 What is the length (in nucleotides) of the wild-type β -globin primary RNA transcript (pre-mRNA) and how many amino acids are present in the wild type β -globin protein?

	Length of β -globin primary RNA transcript	No. of amino acid in wild type β -globin protein
A	1600	146
B	620	146
C	1600	206
D	620	206

- 9 A base-pair substitution mutation (m2) indicated above, destroys the 3' splice site located in the first intron of the β -globin gene. The splice site is the site where the DNA will be cut by enzymes.

Choose the correct explanation as to whether the m2 mutation would result in any change in the length of the primary RNA transcript and mature RNA transcript made from the mutant β -globin gene.

	Pre-mRNA	Mature mRNA
A	No change as this is before the mRNA is spliced.	The splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
B	No change as this is before the mRNA is spliced.	The splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.
C	An increased in its length as this is before RNA splicing.	The splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
D	A decrease in its length as this is before RNA splicing.	The splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.

- 10** Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

first position	second position				third position
	U	C	A	G	
U	phe phe leu leu	ser ser ser ser	tyr tyr STOP STOP	cys cys STOP trp	U C A G
C	leu leu leu leu	pro pro pro pro	his his gln gln	arg arg arg arg	U C A G
A	ile ile ile met	thr thr thr thr	asn asn lys lys	ser ser arg arg	U C A G
G	val val val val	ala ala ala ala	asp asp glu glu	gly gly gly gly	U C A G

Which event occurs to explain the information given above?

- A** There is no tRNA with an anticodon complementary to the first codon.
- B** The first codon is removed from the mRNA transcript in post-transcriptional modification.
- C** The mRNA binds to the rRNA in the second codon position.
- D** The first amino acid on the polypeptide chain is removed in post-translational modification.

- 11 Some bacterial drugs can affect the synthesis of proteins.

Antimicrobial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	Genetic code misread during translation	Prevents bind of tRNA to ribosome

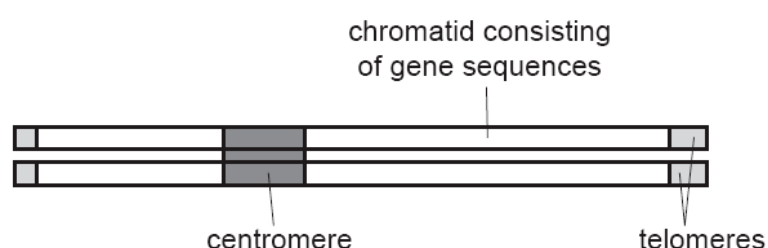
Which is the correct set of immediate effects of these drugs?

Antimicrobial drug	Rifampicin	Streptomycin	Tetracycline
A	Defective protein synthesized	mRNA does not bind to ribosome	Amino acids not added to growing chain
B	Transcription prevented	Defective protein synthesized	Amino acids not added to growing chain
C	Defective protein synthesized	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesized	mRNA does not bind to ribosome

- 12 The ends of eukaryotic chromosome contain a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

The diagram shows a eukaryotic chromosome.



What is a consequence of the loss of repeating DNA sequences from the telomeres?

- A** The cell will begin the synthesis of different proteins.
- B** The cell will begin to differentiate as a result of the altered DNA.
- C** The number of mitotic divisions the cell can make will be limited.
- D** The production of mRNA will be reduced.

- 13 The table describes different kinds of mutations in DNA.

Mutation	Name
From purine to other purine	Transition
From pyrimidine to other pyrimidine	Transition
From purine to pyrimidine	Transversion
From pyrimidine to purine	Transversion

The diagram represents part of a DNA molecule.

G A T A C C A
C T A T G G T

Which diagram shows the DNA molecule with only transversion(s)?

- A**

G T T A T C A
C A A T A G T
- B**

G A A A C A A
C T T T G T T
- C**

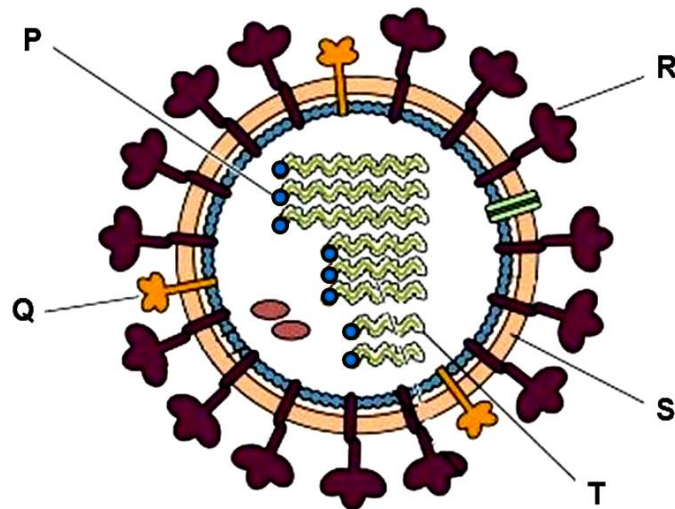
A A T A C C A
T T A T G G T
- D**

G A T A T C A
C T A T A G T

- 14 HIV infections may theoretically be treated by

- A** inhibiting RNA polymerase action in reverse transcriptase.
- B** designing drugs to bind to integrase to prevent transport of genome into nucleus.
- C** preventing protease from degrading host proteins.
- D** infusing stem cells with genes that code for mutant anti-receptors to prevent infection.

- 15 The diagram shows the structure of an influenza virus.

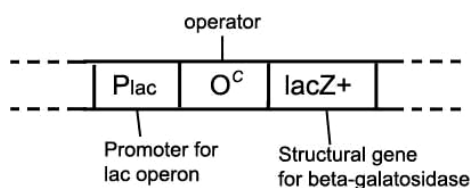


Which of the following statements concerning the lettered components are correct?

- I. Mutations that disrupt the function of **P** will result in the inability of the virus to initiate infection in the host cell.
- II. Influenza viruses are able to cause pandemics in humans due to changes in **Q** and **R** by a slight mutation.
- III. **Q** and **R** are unlikely targets for vaccination because they undergo mutation constantly.
- IV. The symptoms associated with influenza infection are directly due to loss of epithelial cells as the new viruses obtain **S** during budding.
- V. The host cell enzymes are not required to form the complementary RNA from **T**.

- A** I and V
- B** I, II and V
- C** II, III and IV
- D** III, IV and V

- 16** The lac operon of an *E. coli* strain carries the P_{lac} O^c $lacZ^+$ DNA sequence as shown below. In this sequence, the $lacZ$ allele is normal ($lacZ^+$) but the operator (O^c) is mutated. The O^c mutant is unable to bind the lac repressor.



How would the expression of the $lacZ$ gene be affected in this strain of cells?

- A** There will be normal regulation of lactose metabolism.
- B** There will be continuous expression of beta-galactosidase regardless of glucose levels.
- C** The inability to synthesis the $lacZ$ gene product, beta-galactosidase.
- D** The $lacZ$ gene is inducible, but unable to be repressed by high glucose levels.

- 17** The statements give information about genetic inheritance.

- I. At fertilization an organism inherits one set of chromosomes from each parent. Each set carries one of each type of gene.
- II. Each set of chromosomes carries specific genes at specific places.
- III. The same gene may exist with different DNA sequences due to mutation.
- IV. Individuals may have the same DNA sequences for both copies of a gene.
- V. Individuals may have two different DNA sequences of a gene.
- VI. The expression of some versions of genes is hidden by the expression of other versions of the same gene.

Which row matches each statement to the genetic term that it describes?

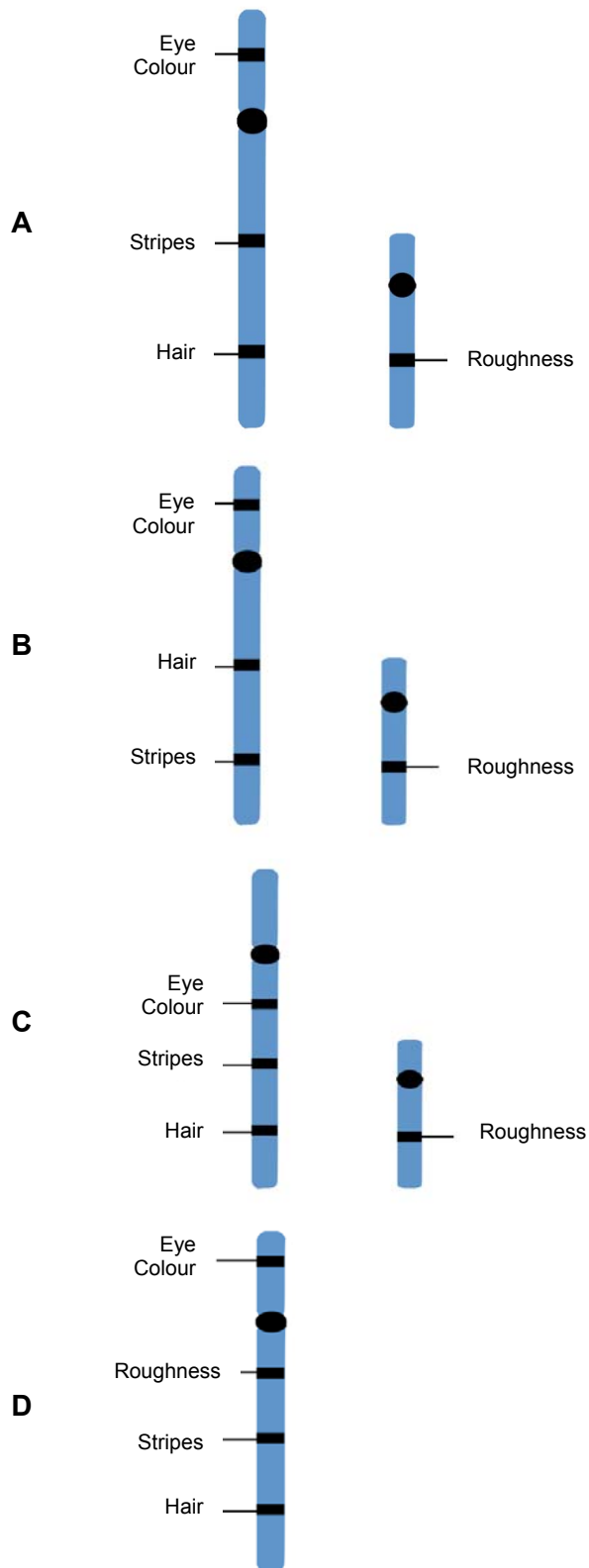
	I	II	III	IV	V	VI
A	alleles	loci	genotype	homozygous	heterozygous	dominant
B	loci	genotype	heterozygous	alleles	homozygous	recessive
C	genotype	loci	alleles	homozygous	heterozygous	recessive
D	loci	homozygous	genotype	alleles	dominant	heterozygous

- 18** Several characteristics – hair on body, stripes on body, eye colour and roughness of body - were investigated in a *Drosophila* to determine the linkage of these genes.

F1 individual of each of the following cross was produced by crossing pure bred parents. Subsequently, a test cross was conducted on one of the F1 to determine the relative distance between the two genes involved.

	Test cross with F1 individual		Number of offspring of each phenotype from test cross			
	F1 individual	Test cross individual				
I	Hairy, striped body	Hairless, plain body	124 hairy, striped body	130 hairless, plain body	22 hairless, striped body	24 hairy, plain body
II	Hairy body, red eye	Hairless body, white eye	80 hairy body, red eye	84 hairless body, white eye	54 hairy body, white eye	56 hairless body, red eye
III	Striped, rough body	Plain, smooth body	72 striped, rough body	74 plain, smooth body	75 striped, smooth body	71 plain, rough body
IV	Striped, body red eye	Plain body, white eye	104 plain body, white eye	112 striped body, red eye	37 striped body, white eye	37 plain body, red eye

Which of the following correctly illustrates the relative position of these four genes?



- 19 Test crossing of some tomato plants gave the following results.

Plants with round leaves and are tall	157
Plants with round leaves and are dwarf	143
Plants with oval leaves and are tall	140
Plants with oval leaves and are dwarf	160

The expected phenotypic ratio from such test cross is 1 : 1 : 1 : 1. A χ^2 (chi-squared) test was performed to test the significance of the difference between the observed and expected results. The calculated $\chi^2 = 8.36$.

distribution of χ^2			
number of degrees of freedom (v)	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

Which combination correctly describes the result of the χ^2 test?

	Number of degrees of freedom (v)	probability	Significance of difference between observed and expected results
A	3	< 0.05	Not significant
B	3	< 0.05	Significant
C	4	> 0.05	Not significant
D	4	> 0.05	significant

- 20** The diagram below shows the ears of maize (*Zea mays*). On the left, is variety Tom Thumb, Black Mexican on the right, and that of their F₁ hybrid in the centre.



(photo by John Doebley)

A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length of both parents, the F₁ and the F₂ generations, was measured in centimetre. This is given in the table below with the number of ears in each length category (e.g. 14 of the F₁ plants produced ears 12 cm in length).

	Ear length (cm)																
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Black Mexican parents									3	11	12	15	26	15	10	7	2
Tom Thumb parents	4	21	24	8													
F₁					1	12	12	14	17	9	4						
F₂			1	10	19	26	47	73	68	68	39	25	15	9	1		

Which of the following statement is incorrect?

- A** Ear length exhibits continuous variation as a result of additive effect of different genes.
- B** Difference in ranges of ear length between parental varieties and offspring is due to environmental factors.
- C** Environmental factors smooth out the differences between phenotypic groups within the parental varieties and offspring.
- D** F_2 has a greater range of ear length as compared to F_1 due to crossing over, independent assortment and segregation of genes.

21 If a non-disjunction of chromosomes occurred in the first meiotic division, and one of the resulting gamete fertilizes with a normal gamete, which of the following zygote(s) may possibly be formed?

- I. Normal zygote
- II. Monosomic zygote
- III. Trisomic zygote

- A** III only
- B** II and III only
- C** I and II only
- D** I, II and III

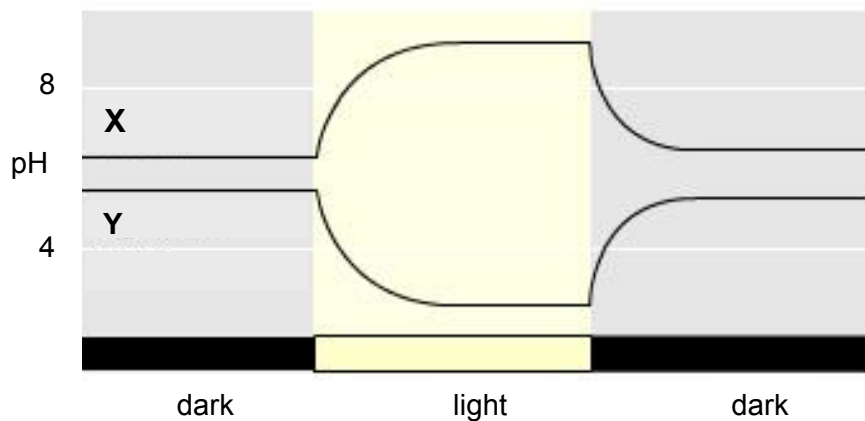
22 Which feature of the life cycle of some viruses may result in the development of cancer?

- A** Viral RNA can integrate into the chromosomes of host cells.
- B** Viruses can cause cell lysis and spread to other host cells.
- C** Viruses can cause loss of function mutations in proto-oncogenes.
- D** Viruses can increase the rate of the cell cycle of host cells.

23 Which of the following shows correctly the types of genetic changes that are least likely to be found in an oncogene and a tumor suppressor gene of tumor cells?

	Oncogene	Tumor suppressor gene
A	Gene amplification	Chromosomal deletion
B	Chromosomal addition	Substitution mutation
C	Substitution mutation	Chromosomal translocation
D	Mutation resulting in a stop codon	Gene amplification

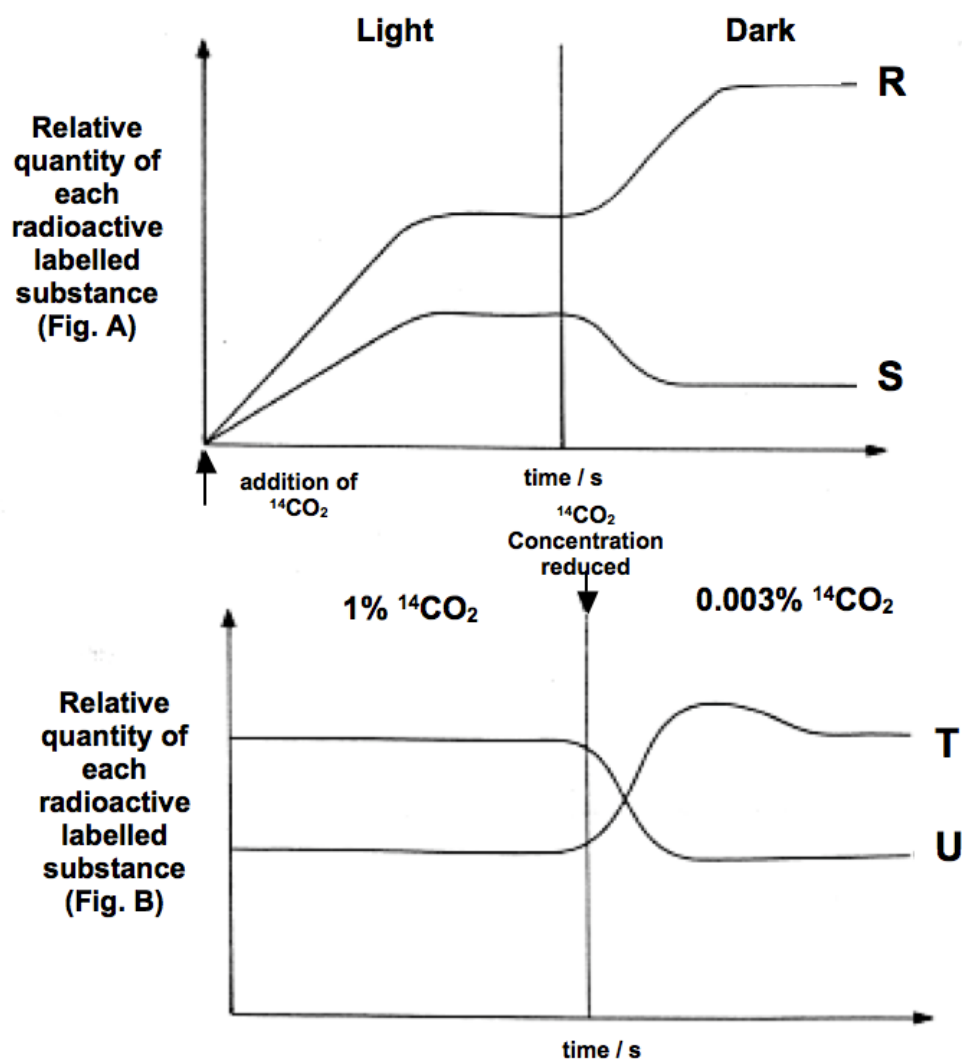
- 24** The diagram below consists of two graphs showing the changes in pH of two structures **X** and **Y**, of a chloroplast in the absence and presence of light.



Which of the following correctly identifies structures **X** and **Y**?

	X	Y	Reason
A	Thylakoid space	Stroma	Protons are pumped from the thylakoid space into the stroma, hence decreasing the latter's pH.
B	Stroma	Thylakoid space	Protons are pumped from the stroma into the thylakoid space hence decreasing the latter's pH.
C	Thylakoid membrane	Stroma	Protons are pumped from the thylakoid membrane into the stroma, hence increasing the latter's pH.
D	Stroma	Thylakoid space	Protons are pumped into the stroma from the thylakoid space, hence increasing the latter's pH.

- 25 Fig. A and Fig. B show the results of two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and the radioactive isotope ^{14}C to label carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.



With reference to Fig. A and Fig. B, identify the graphs **R**, **S**, **T** and **U**.

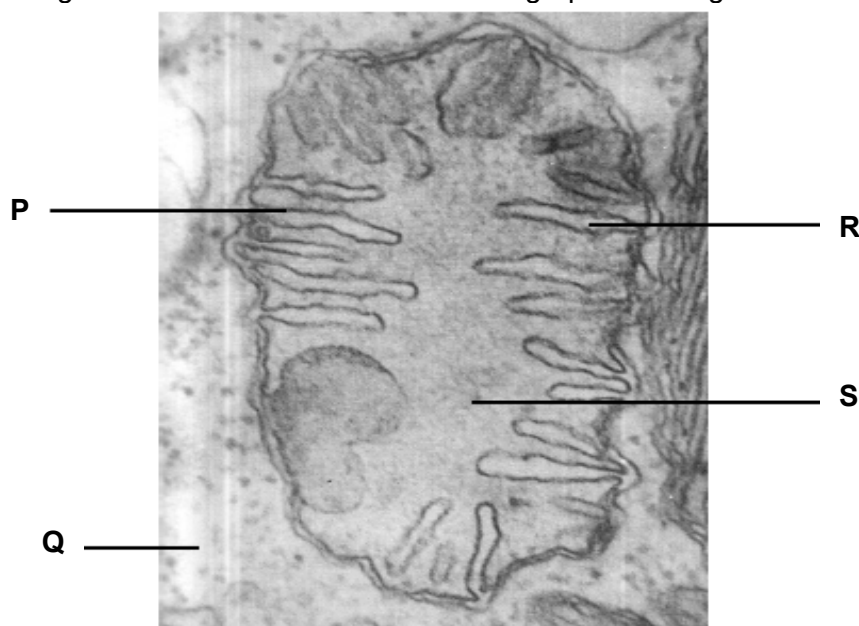
	R	S	T	U
A	RuBP	GP/PGA	GP/PGA	RuBP
B	GP/PGA	RuBP	GP/PGA	RuBP
C	GP/PGA	RuBP	RuBP	GP/PGA
D	RuBP	GP/PGA	RuBP	GP/PGA

- 26** 2,4-dinitrophenol is a chemical that targets the inner mitochondrial membrane to uncouple oxidative phosphorylation from electron transport. It allows protons to cross the inner mitochondrial membrane and thus dissipates the proton gradient.

Which of the following is true of a cell treated with 2,4-dinitrophenol?

	Ability to use glucose	Ability to use oxygen	Ability to produce carbon dioxide	ATP yield
A	Yes	Yes	No	Decreases
B	Yes	Yes	Yes	Decreases
C	Yes	No	Yes	Increases
D	No	Yes	No	Increases

- 27** The figure below shows an electron micrograph of an organelle.



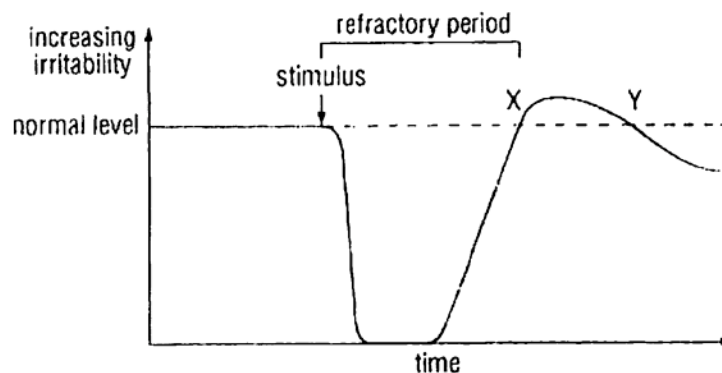
Match the following processes with the structures labelled **P – S** above.

	Breakdown of fructose-6-phosphate	Oxidative phosphorylation only	Temporary lowering of pH	Formation of water
A	Q	R	S	P
B	R	S	P	Q
C	S	P	Q	R
D	Q	P	R	S

- 28 What are the effects that insulin will bring about when it binds to its target cells?

	Glycogenolysis	Glycogenesis	Gluconeogenesis
A	Increase	Increase	Increase
B	Increase	Decrease	Increase
C	Decrease	Increase	Decrease
D	Decrease	Decrease	Decrease

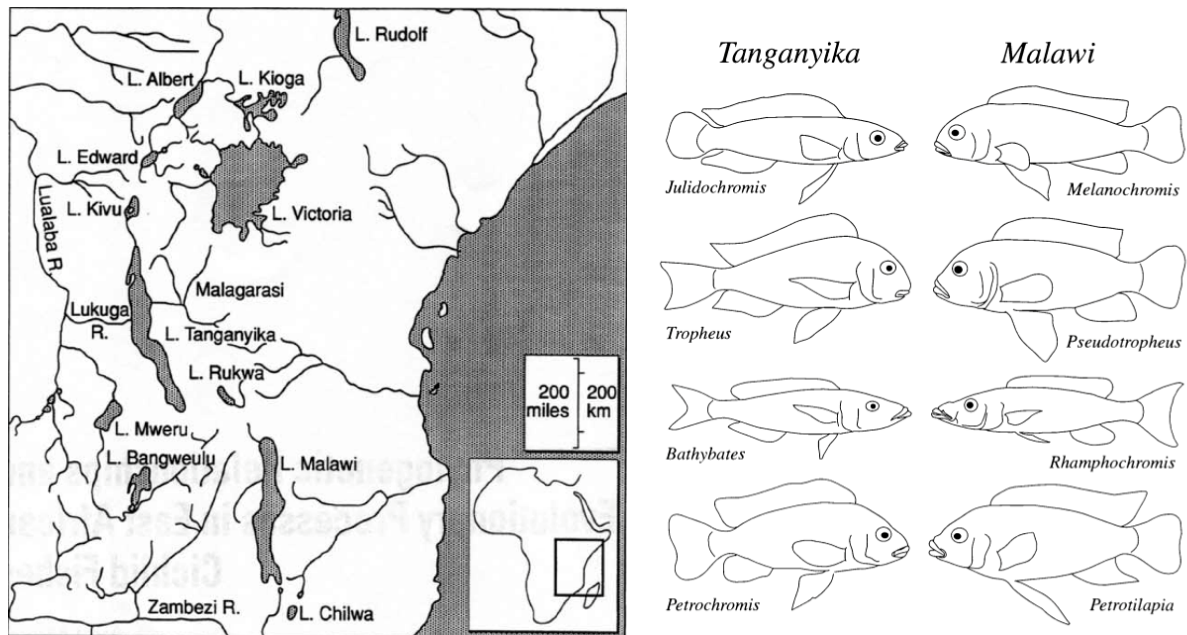
- 29 The figure below shows the relative irritability of a stimulated nerve fibre.



What would happen if a second stimulus is applied during period XY?

- A** The action potential will take a longer time to be generated.
B The reaction to the stimulus will be greater than normal.
C The nerve fibre will respond repeatedly to a single stimulus.
D A stimulus below the normal threshold will cause an action potential.
- 30 What is the correct sequence of events in the signal transduction pathway when insulin interacts with receptor tyrosine kinases?
1. Activation of tyrosine kinase in tails
 2. Dimerization of receptor polypeptides
 3. Autophosphorylation of tyrosine molecules on the cytoplasmic tail
 4. Activation of series of relay proteins
- A** 1 2 3 4
B 2 3 1 4
C 2 1 3 4
D 1 3 2 4

- 31 A study was done on the cichlids in two East African lakes far apart from each other as shown in the figure below.



Which would best help distinguish between an explanation based on convergent evolution versus one based on homology?

- A Genes coding for body shape appears at a different gene locus across different species.
 - B The species share a common ancestor.
 - C The species have almost identical nucleotide sequence that code for these proteins.
 - D All species are well adapted to their particular environment.
- 32 On the tiny Lord Howe Island, 600 miles east of Australia, there are two species of palm which seem, from DNA analysis, to be descended from one original species.

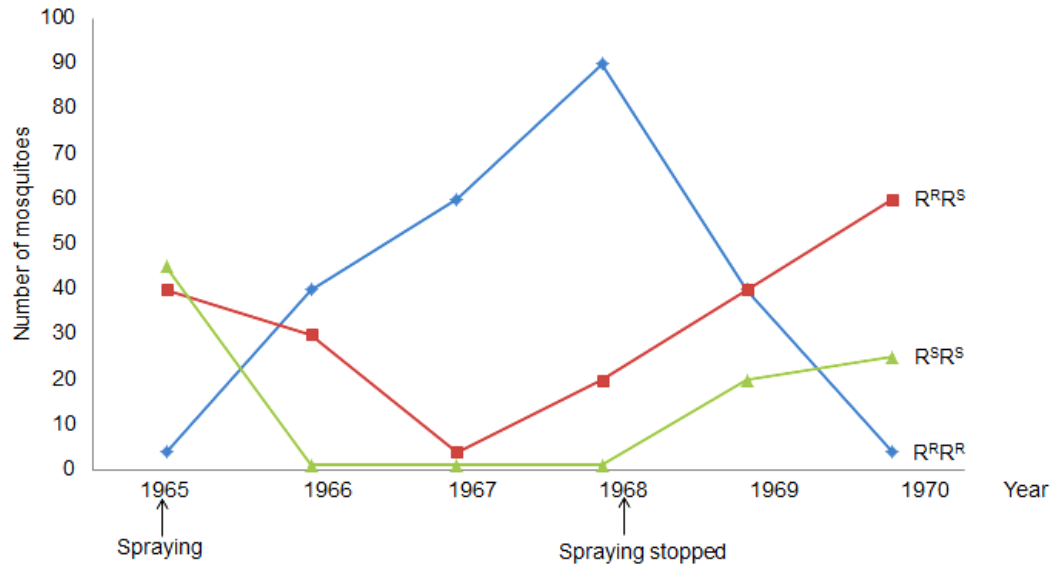
Factors involved in this speciation on this tiny island include:

- I. linkage of genes for soil tolerance and flowering time
- II. variation in flowering time
- III. variation of soil tolerance
- IV. variation of soil types on the island

What is the correct sequence to explain this speciation?

- A I -> II -> III -> IV
- B II -> I -> IV -> III
- C III -> IV -> I -> II
- D IV -> III -> II -> I

- 33** In the mosquito, there is a gene locus which has two alleles, R^R = Resistance and R^S = Sensitive, which are involved in resistance, in particular to DDT, the insecticide. The graph shows the number of mosquitoes of the three phenotypes (and genotypes), collected in 1965, when DDT was first used, through to 1970, two years after the spraying of DDT stopped.



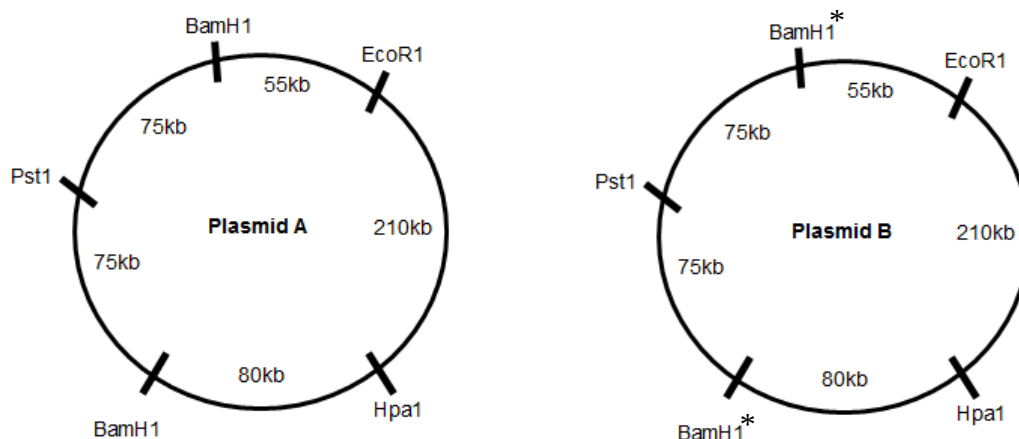
From the data, it is possible to conclude that

- A** the frequency of R^S allele is greater than the frequency of R^R allele in 1965.
- B** many generations after the removal of DDT the R^R allele would disappear from the population
- C** after the removal of DDT from the environment in 1968, having the $R^R R^R$ genotype reduces the chances of survive.
- D** between 1967 and 1968 in the presence of DDT in the environment, the mosquitoes with $R^R R^S$ genotype are most likely to survive.

34 Which combination of definitions is correct?

	taxonomic group	phylogeny	binomial
A	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the names of the family and genus of the organism
B	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the family and genus of the organism
C	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the genus and species of the organism
D	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the names of the genus and species of the organism

- 35 Figure below shows 2 plasmids (A and B) and the respective restriction sites of various restriction enzymes. In plasmid B, mutations were deliberately introduced and affected restriction sites were denoted by *. The lengths of the plasmid DNA between consecutive restriction sites are also indicated.



He carried out a series of 2 experiments by adding different restriction enzymes to the plasmids. Each tube was then left to incubate at 37°C for about 15 minutes (sufficient time for complete digestion to take place). The results are shown in table below.

Tube number	Components	Fragment sizes / kb
1	Plasmid A + 2 restriction enzymes	80, 150, 265
2	Plasmid B + 2 restriction enzymes	155, 340

Identify the enzymes that were added to tubes 1 and 2.

	Tube 1	Tube 2
A	HpaI and EcoRI	EcoRI and HpaI
B	BamHI and PstI	PstI and EcoRI
C	BamHI and HpaI	HpaI and PstI
D	BamHI and EcoRI	HpaI and BamHI

36 Which of the following processes of PCR and DNA replication is not true?

	PCR	DNA replication
A	RNA primers were synthesized synthetically.	RNA primers were synthesized by primase.
B	Taq polymerase synthesises the daughter strand in the 5' to 3' direction.	DNA polymerase III synthesises the daughter strand in the 5' to 3' direction.
C	Taq polymerase does not contain 3' to 5' exonuclease.	DNA polymerase III contains 3' to 5' exonuclease.
D	DNA strands separate after the breaking of hydrogen bonds after the temperature was raised to 90oC	DNA strands separate after the breaking of hydrogen bonds by helicase

37 Some of the goals of the Human Genome Project are listed.

- I. to determine the base sequence of the whole of the human genome
- II. to identify the genes encoded by the base sequence
- III. to find the locations of the genes on the 23 pairs of human chromosomes

Which goals had to be achieved to allow the development of a single-stranded DNA probe for detecting the presence of a mutant allele on human chromosome 7?

	I	II	III
A	✓	✓	
B	✓		✓
C	✓	✓	✓
D		✓	✓

- 38** Seven skeletons were found in an unidentified grave. To establish the relationship between these seven individuals, small amount of DNA were isolated from these skeletons and polymerase chain reaction (PCR) was carried out. Genetic fingerprinting was then carried out on these DNA.

The chart shows some of the results obtained from the genetic fingerprinting of seven of the skeletons, three children and four adults.

Child 1	Child 2	Child 3	Adult 1	Adult 2	Adult 3	Adult 4
		—			—	
—	—				—	—
—	—	—	—	—		—
—	—	—		—	—	
—		—	—	—		—
—	—	—	—	—	—	—
—	—	—	—	—	—	
—	—	—	—	—	—	
—	—	—	—	—	—	—
				—		
						—
—		—	—		—	
	—	—			—	

All the children on the chart had the same parents. Which of the following show the children's parents?

- A** Adult 1 and Adult 2
- B** Adult 1 and Adult 3
- C** Adult 2 and Adult 4
- D** Adult 2 and Adult 3

39 Gene therapy is the insertion of a therapeutic gene into a target cell with dysfunctional genes. Which of the following are techniques of gene therapy to treat a genetic disease caused by a gain-of-function mutation?

- I. A therapeutic allele with a promoter is inserted into a non-specific location within the genome
- II. A therapeutic allele with a promoter is inserted in a specific location upstream of the dysfunctional gene
- III. A therapeutic allele with a promoter enters the cell and exist as an circular extrachromosomal plasmid
- IV. An allele complementary to the normal allele with a promoter is inserted into a non-specific location within the genome

- A** I, II and III only
- B** II only
- C** IV only
- D** II and IV only

40 Some plant tissue culture techniques involve the production of protoplasts using leaf tissue. The tissue is treated to obtain isolated, structurally altered cells. The treatment involves incubation in a solution that contains enzymes such as cellulose and pectinase.

Which statement about protoplast is not correct?

- A** Protoplasts are pluripotent and regenerate into whole plants when provided with the correct growth factors.
- B** Protoplasts are more susceptible to microbial contamination.
- C** Protoplasts are used for the production of genetically engineered plants as they take up naked DNA easily.
- D** Protoplasts need to be maintained in the solution of the same water potential to prevent lysis.

~THE END~